If a programming practice or tool is successful, following or using it will make it easier to

- add the code you need in the short term.
- continue to add code to the project over time.

All kinds of gems in the Ruby and Rails ecosystem help with the first goal (including Rails itself). Testing is normally thought of as working toward the second goal. That's true, but often people assume the only contribution testing makes toward long-term application health is verification of application logic and prevention of regressions. In fact, over the long term, test-driven development tends to pay off as good tests lead toward modular designs.

Cost and Value

Tests have costs and tests have value. The goal of testing is to minimize the cost of doing testing and maximize the value. This leads to the question of what the costs are and what the value is.

The cost is easier to see and define. The cost of a test includes the following:

- The time it takes to write the test
- The time it takes to run the test every time the suite runs
- The time it takes to understand the test
- The time it takes to fix the test if it breaks and the root code is okay
- In some cases, the time it takes to change the code to make it testable

The currency I'm using here is time. Tests cost time. Eventually every programming decision costs time. When people talk about code quality making it easier or harder to make changes, they are inevitably talking about the amount of time it takes to change the code.

How can writing a test save time? Here's an incomplete list:

- The act of writing the test makes it easier to define the structure of the code.
- Running an automated test is frequently faster than manually going through elaborate integration steps.
- The test can provide efficient proof that code is working as desired.
- The test can provide warning that a change in code has triggered a change in behavior.
- The test can make it easier to locate and fix bugs.

Both the cost and the value of the test are scattered over the entire life cycle of the code. The cost starts in a burst when the test is written, but continues